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2015.0 RANGE ROVER (LG), 303-03

ENGINE COOLING - TDV8 4.4L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the engine cooling system and operation, refer to the relevant Description and Operation section in the Workshop Manual.

REFER to: [Engine Cooling](#) (303-03F Engine Cooling - TDV8 4.4L Diesel, Description and Operation).

INSPECTION AND VERIFICATION



CAUTION:

Diagnosis by substitution from a donor vehicle is NOT acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

1. Verify the customer concern.
1. Visually inspect for obvious mechanical or electrical faults.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Coolant leaks ▪ Coolant hoses ▪ Coolant expansion tank ▪ Radiator ▪ Heater core ▪ Active grill air shutter ▪ Accessory drive belt ▪ Cooling fan	▪ Fuses ▪ Harnesses ▪ Loose or corroded connector(s) ▪ Engine coolant temperature sensor ▪ Engine oil temperature sensor

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Coolant loss	▪ Hoses ▪ Hose connections ▪ Radiator ▪ Water pump ▪ Heater core ▪ Gaskets ▪ Engine casting cracks ▪ Engine block core plugs	Carry out a visual inspection. If there are no obvious leaks, carry out a cooling system pressure test. Rectify any leaks as necessary. Refer to the relevant section of the workshop manual
Overheating	▪ Low /contaminated coolant ▪ Thermostat ▪ Cooling fan ▪ Engine coolant temperature sensor ▪ Restricted air flow over the radiator ▪ Active grill air shutter	Check the coolant level and condition. Carry out a cooling system pressure test. Rectify any leaks as necessary. Check the thermostat and rectify as necessary. Check the cooling fan operation, make sure the cooling fan rotates freely. Check for obstructions to the air flow over the radiator. Check the active grill air shutter operation using the manufacturer approved service tool
Engine not reaching normal temperature	▪ Thermostat ▪ Cooling fan	Check the thermostat operation. Check the cooling fan operation, make sure the cooling fan is not seized

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: Diagnostic Trouble Code (DTC) Index - TDV8 4.4L Diesel, DTC: Module Name: Engine Control Module (100-00, Description and Operation).